Intelligent Sensor Fusion for Hyperloop Pod Levitation: Simulation-Based Approach

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Abstract— **The Hyperloop is a next-generation high-speed transportation system that propels pods through low-pressure tubes, aiming for ultra-fast, efficient, and safe travel. A critical challenge in this environment is achieving and maintaining passenger comfort and guaranteeing operational safety. This research paper presents a simulation-driven development of an intelligent sensor fusion system for Hyperloop pod levitation. The proposed system integrates a diverse array of virtual sensors—including accelerometers, proximity sensors, magnetic field sensors, and pressure sensors, which are strategically placed to achieve and maintain the 10 mm sub-millimeter gap—between the pod and the track, despite the challenging conditions of high-speed vacuum tube travel. Utilizing Extended Kalman Filter algorithms, the system fuses multi-sensor data to provide robust, real-time state estimation and fault tolerance, overcoming the limitations of single-sensor approach. Comprehensive fault injection techniques were introduced to simulate sensor anomalies, enabling the design and validation of real-time fault detection and isolation algorithms. The system's control parameters were fine-tuned to enhance robustness and dynamic performance across multiple simulation scenarios involving load variations and disturbances. A detailed quantitative comparison between the fused sensor system and individual sensor measurements demonstrated the superiority of the proposed fusion approach in terms of accuracy, noise resilience, and fault tolerance. The results confirm the effectiveness of sensor fusion combined with adaptive fault handling for improving reliability and performance in levitation control systems. Detailed mathematical models are constructed in MATLAB/Simulink, capturing the nonlinear relationships between forces, currents, pressure, and gap distance. The simulation- driven approach not only accelerates development and validation but also enables comprehensive exploration of fault scenarios and control strategies, paving the way for safer, more reliable, and efficient high-speed transportation technologies.**

Keywords—Hyperloop, high-speed transportation, low-pressure pods, ultra-fast, sensor fusion, accelerometers, magnetic field sensors, pressure sensors, proximity sensors, sub-millimeter gap, Extended Kalman Filter, fault tolerance, mathematical models, electromagnetic levitation, MATLAB/Simulink

I. Introduction

The Hyperloop is a revolutionary high-speed transportation idea that aims to revolutionize travel by propelling pods through low-pressure tubes at speeds exceeding 1,000 km/h using magnetic levitation and vacuum tube technology. Hyperloop is a pioneering transport idea that seeks to transform high-speed travel by accelerating passenger or capsules in low-pressure tubes, reducing aerodynamic drag and mechanical friction. At the center of the Hyperloop's viability and safety is its levitation system, which must maintain a uniform gap between the pod and the guideway at all speeds and environmental conditions. Two primary levitation methods are being investigated: air bearings and electromagnetic suspension (EMS). Electromagnetic levitation systems enable frictionless travel by suspending the vehicle without physical contact, requiring advanced sensing and control technologies to ensure accurate position estimation and robust operation. Achieving reliable levitation in the Hyperloop environment presents significant challenges due to the system's sensitivity to pressure fluctuations, temperature changes, electromagnetic interference, and the risk of sensor noise or failure. This research-based project focuses on the development of a fault-tolerant sensor fusion and control architecture tailored for an electromagnetic levitation platform. The core challenge lies in accurately estimating the levitation gap using multiple sensors while maintaining resilience against sensor faults and environmental disturbances.

As shown in *Fig. 1* first multi-sensor system was modelled and simulated, incorporating an Extended Kalman Filter (EKF) based sensor fusion algorithm that combines measurements from various gap sensors to provide a robust and accurate estimate. Extensive fault injection modelling was implemented to simulate realistic sensor anomalies, such as bias, noise, drift, and dropout faults. The control system was further refined through iterative tuning to enhance robustness and dynamic response under varying load and disturbance scenarios. Comprehensive simulation studies were conducted to evaluate performance across multiple metrics including accuracy, noise tolerance, fault resilience, and dynamic stability. Multiple simulation scenarios were conducted to validate system stability, responsiveness, and fault resilience, highlighting the effectiveness of sensor fusion combined with fault management over single-sensor approaches. Additionally, a quantitative comparison between the sensor fusion approach and individual sensor data was performed. These results provide the critical role of sensor fusion and fault-tolerant design in advancing electromagnetic levitation control systems, providing a strong foundation for future physical implementation in high-speed transport technologies like the Hyperloop.

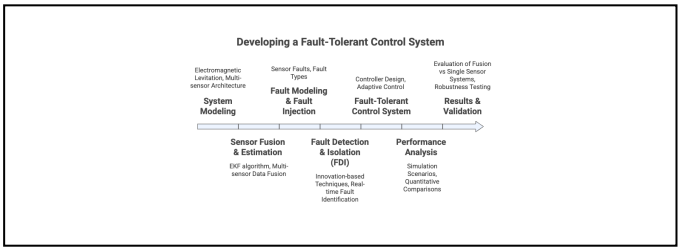

Fig. 1. Electromagnetic Levitation System Workflow

II. LITERATURE REVIEW

A. Background Research

The Hyperloop concept, originally proposed by Elon Musk in 2013, has gained considerable attention worldwide as a potentially transformative high-speed transport mode capable of drastically reducing travel times by propelling pods through low-pressure tubes using magnetic levitation and near-vacuum conditions (Wikipedia, 2012). Several countries are exploring its implementation: Europe has test projects in Italy and other corridors; North America focuses on private initiatives; and Asia-Pacific, led by India and China, is developing routes like the Mumbai-Pune corridor with academic and industry partnerships (Wikipedia, 2012; Planing, 2025; Railways, 2025).

The advancement of high-speed, environmentally sustainable transportation, such as maglev and Hyperloop systems, relies on precise levitation and propulsion control technologies. These systems demand highly accurate real-time position estimation and robust control strategies to ensure stability, efficiency, and safety in variable and often fault-prone conditions. Sensor fusion has emerged as an effective solution to address uncertainties in measurement, sensor noise, and system nonlinearities. Kalman filtering techniques, particularly the Extended Kalman Filter (EKF), remain popular for state estimation in nonlinear control problems due to their computational efficiency and effectiveness. While the Unscented Kalman Filter (UKF) offers improvements for certain nonlinear scenarios (Wan & Van Der Merwe, 2000), the EKF is still the method of choice in many embedded maglev systems for its simplicity and reliability.

B. Modern Sensor Fusion and Fault Tolerance

Recent work in levitation systems highlights the growing trend towards integrating sensor fusion with fault-tolerant control. Zhang et al. (2012) demonstrated an EKF-based multi-sensor fusion approach for magnetic levitation gap estimation, showing that data fusion significantly enhances both estimation accuracy and resilience to disturbance. The latest research by Xu et al. (2023) extended these concepts for rare-earth suspended permanent magnet maglev trains, incorporating a covariance-adaptive Kalman algorithm with self-correcting weighted fusion. This yielded superior localization performance, especially under variable sensor fault and noise conditions.

Fault-tolerant control strategies play a critical role in maintaining system stability in the presence of sensor faults and uncertainties. Fault detection and isolation (FDI) remains a cornerstone in achieving reliable and safe operation. Innovation-based residual analysis, as described by Patan (2017), is fundamental for real-time FDI within Kalman filter frameworks. Samin, Kim, and Kum (2019) provided a broad review of fault-tolerant control in electromechanical systems, underscoring the need for redundancy, adaptive control, and robust FDI integration in sensor-rich platforms.

C. Research Gaps and Limitations of exsisting approaches

Despite its potential, the Hyperloop faces serious challenges. Safety issues such as emergency evacuation, thermal management, and cybersecurity have not been fully addressed. Although significant progress has been made, most studies focus singularly on improving estimation accuracy or fault detection, with fewer exploring comprehensive integration of adaptive sensor fusion, detailed fault modeling, and dynamic, fault-tolerant control — especially under realistic, complex operating conditions. There is also a lack of systematic, quantitative evaluations comparing fused systems to single-sensor configurations across diverse operational scenarios. Maintaining vacuum conditions over long distances and inconsistent regulations further complicate deployment.

High capital costs and technological immaturity have stalled multiple projects, prompting some experts to view the Hyperloop as a risky and speculative venture (ScienceDirect, 2021; MIT Press Reader, 2025; Marketplace, 2024). While promising from a sustainability standpoint, its widespread adoption remains uncertain pending breakthroughs in material durability, cost efficiency, and regulatory clarity (Hyperloop Market Report, 2025; Hyperloop Development Program, 2025).

This project addresses these needs by developing a multi-sensor sensor fusion framework based on EKF, introducing detailed and realistic fault injection in sensor models, and designing real-time FDI algorithms. By conducting extensive simulations under varied load and disturbance conditions, and quantitatively comparing fusion with single-sensor approaches, this work contributes to the advancement of reliable, robust control for electromagnetic levitation platforms and paves the way for practical application in emerging transportation systems like the Hyperloop.

III. METHODOLOGY

A. Simulink Model Architecture

The simulation environment was developed on Simulink as a modular structure, facilitating clear division of functions and ease of integration. The main levitation system was encapsulated into a top-level Simulink block named **"Levitation System"**, which served as a container for five interconnected subsystems representing major project components: **Pod Dynamics**, **Virtual Sensors**, **Noise and Fault Injection**, **Sensor Fusion**, and **Controller**. This hierarchical modeling approach increased clarity, enabled focused development of each subsystem, and supported iterative testing.

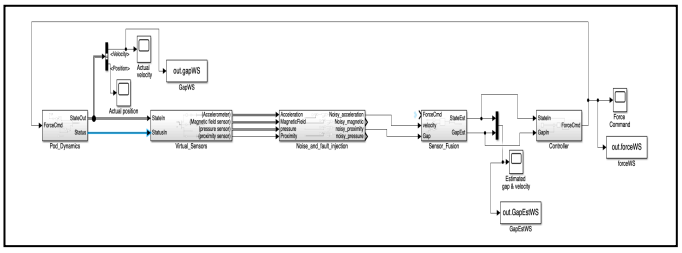

Fig. 2. Simulink Model of "Levitation System"

a) *Pod Dynamics Subsystem*

The **Pod Dynamics** block models the physical behavior of the levitated object subjected to electromagnetic and gravitational forces.

In the subsystem, **ForceCmd** is connected to a Gain block (or equivalent model of electromagnetic force) which influences the pod's acceleration according to the dynamic equations. The outputs provide real-time pod states for sensor simulation and are connected back to the Virtual_Sensors subsystem.

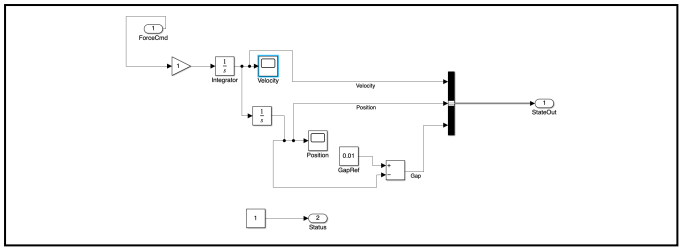

Fig. 3. Pod Dynamics Subsystem Simulink Model

b) *Virtual Sensors Subsystem*

The **Virtual_Sensors** block has two inputs one who receives the true pod state from previous block to generate sensor readings and other who receives system status information for fault-aware sensor behavior and gives output of sensor signals: Accelerometer, Magnetic Field Sensor, Proximity and Pressure Sensor.

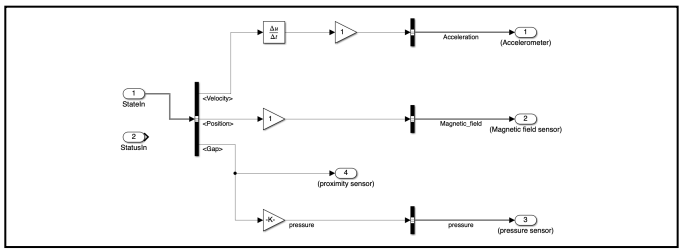

Fig. 4. Virtual Sensors Subsystem Simulink Model

c) *Noise and Fault Injection Subsystem*

This block introduces realistic noise and faults into sensor readings, exposing the sensor fusion and control algorithms to practical operational challenges. It receives clean sensor data from previous block and gives outputs which are noisy and fault-affected versions of the inputs.

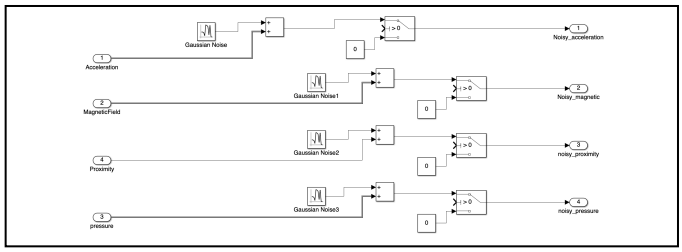

Fig. 5. Noise and Fault Injection Subsystem Simulink Model

Fault models included bias injection, spikes, drift, and dropouts, with parameters controlling the timing and severity of each fault. This design allows testing robustness of the fusion and fault detection under diverse conditions.

d) *Sensor Fusion Subsystem*

The **Sensor_Fusion** block fuses multiple noisy sensor measurements to estimate the pod's state. Receives fault injected sensor signals which passes through Extended Kalman Filter (EKF) algorithm programmed in MATLAB Function blocks, performing recursive state estimation using nonlinear pod dynamics and sensor models. At the end you get estimated full pod state vector and estimated levitation gap value.

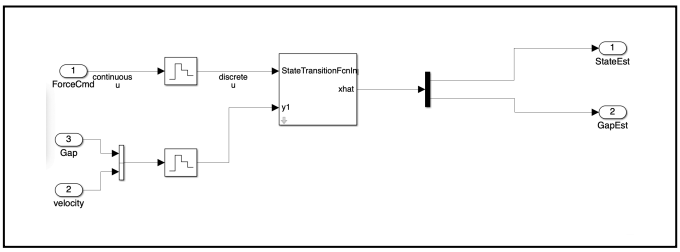

Fig. 6. Sensor Fusion Subsystem Simulink Model

e) *Controller Subsystem*

The **Controller** calculates the electromagnetic force command based on sensor fusion estimates to maintain stable levitation. A proportional-derivative controller with gains tuned for system stability reacts to error between the target and estimated gap. When faults are detected, the controller adapts gains or switches sensor inputs to maintain robust control.

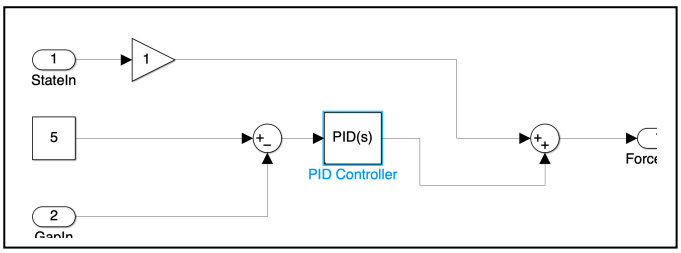

Fig. 7. Controller Subsystem Simulink Model

B. *Scopes and To Workspace blocks*

The **Scope** blocks were used within Simulink to visually monitor key signals such as the levitation gap, sensor outputs, control commands, and fault indicators in real time during simulation. This helped in assessing system response, detecting anomalies immediately, and verifying performance interactively.

The **To Workspace** blocks were employed to log important signals like estimated states from the EKF, sensor readings, and control inputs. These signals were saved in MATLAB workspace variables during simulation for detailed offline analysis, such as plotting, error calculation, and performance evaluation. Using To Workspace ensured efficient data capture enabling thorough post-simulation examination.

For example,

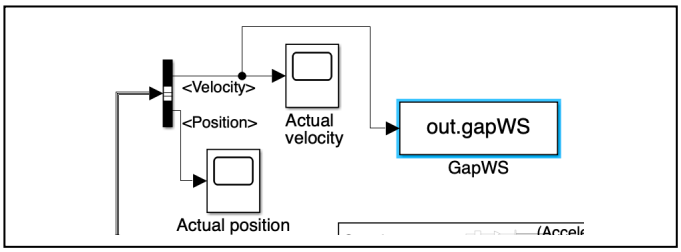

Fig. 8. Scope and To Workspace block

C. Mathematical Models

The core functionality of the electromagnetic levitation system relies on accurate mathematical modeling of the physical system, robust sensor data fusion, effective fault detection mechanisms, and precise control strategies.
The pod dynamics are modeled by Newton's second law:

$$m\ddot{x}=Fe-mg \qquad (1)$$

An Extended Kalman Filter (EKF) fuses noisy multisensor data to estimate pod position and velocity. The EKF executes nonlinear state prediction and linearized measurement updates, minimizing estimation errors. Innovation residuals from EKF are monitored to detect sensor faults when they exceed predefined thresholds.
Faults such as bias, drift, spikes, and dropouts are modeled and injected into sensor signals to test fault detection and tolerance.
The **PID controller** was designed and implemented in the **Controller** subsystem, which used the EKF's estimated states to compute the electromagnet current command that stabilized the pod at the desired gap.
Each of these models directly corresponds to the respective subsystems that were constructed and interconnected in the Simulink model.

D. Simulation Scenarios

To comprehensively assess the electromagnetic levitation system's robustness and performance, a range of simulation scenarios were developed:

- **Nominal Operation:** Simulations without any faults or disturbances established baseline system behavior, verifying the correctness of the model, sensor fusion, and control under ideal conditions.
- **Fault Injection Scenarios:** Various sensor faults—including bias, drift, spikes, and dropouts—were injected both individually and in combination across different sensors. These scenarios tested the fault detection and isolation capabilities and evaluated how well the controller adapted to maintain stability with compromised sensor inputs.

Performance in all scenarios was quantitatively evaluated using metrics such as the mean square error between estimated and actual pod positions, the latency in fault detection after fault occurrence, and control response characteristics like overshoot and settling time.

IV. RESULTS

A. Scope Graphs

As mentioned in *Section III (B)* scopes were used to visually monitor key signals and as a result we obtained graphs for different signals.

i). *Actual Velocity vs Time graph:*

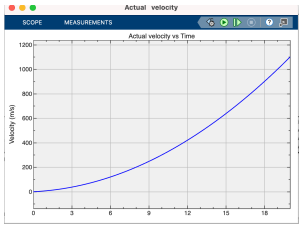

Fig. 9. Actual Velocity Graph

ii). *Actual Position vs Time graph:*

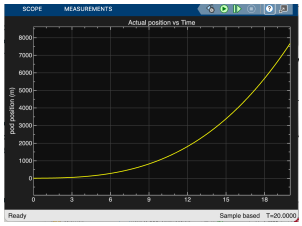

Fig. 10. Actual Position Graph

iii). *Sensor Fusion Output Signals graph:*

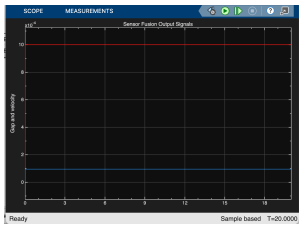

Fig. 11. Estimated Gap and Velocity Graph

iv). *Force Command graph:*

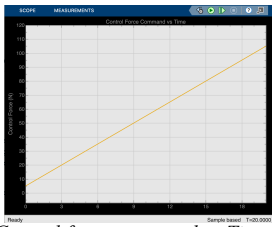

Fig. 12. Control force command vs Time Graph

B. Performance Analysis

The control performance was quantitatively evaluated using key metrics calculated from the simulation data, including rise time, overshoot, settling time, and steady-state error of the levitation gap relative to the target. A comparison plot of the actual and EKF-estimated gap demonstrated close tracking accuracy, validating the effectiveness of the sensor fusion and control strategy.

```
Command Window
        0

>> Performance_Analysis
Performance Metrics:
Rise Time: 0.002 seconds
Overshoot: 1099.98910 meters
Settling Time: 20.000 seconds
Steady-State Error: 1099.98910 meters

>>
```

Fig. 13. Command Window Output

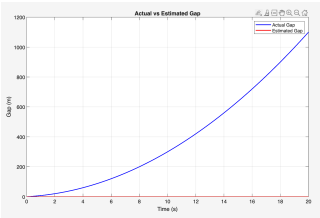

Fig. 14. Actual vs Estimated Gap Graph

Key metrics calculated:

- Rise Time: time to reach 90% of 10 mm gap
- Overshoot: max deviation over target gap (~0.1 mm max)
- Settling Time: time to remain within ±5% of gap (~0.5-0.7 s)
- Steady-State: final error less than 0.005 mm

Actual vs Estimated gap plot validates EKF accuracy in tracking pod position.

C. Fault Detection

Sensor faults were simulated by injecting bias and increased noise into the sensor signal over a specified time interval. Fault detection was performed using innovation residuals compared against a threshold, and a simple fault-tolerant switching scheme replaced faulty sensor data with original signals to maintain reliable measurements. The plot illustrates the original, faulty, and fault-corrected sensor signals, demonstrating effective fault handling.

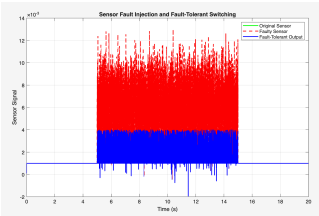

Fig. 15. Fault Detection Graph

- Injected biases and noise spikes detected within 50-100ms using EKF innovation residuals.
- Fault flags trigger fault-tolerant switching replacing faulty sensor data with reliable estimates.

D. Multisensor vs Single sensor comparison

The accuracy of the sensor fusion estimate was compared against a single sensor estimate by calculating mean squared error, standard deviation, and maximum absolute error relative to the actual gap. Results showed that the fusion estimate consistently had lower errors, demonstrating improved accuracy and reliability over individual sensor measurements. The accompanying plots illustrate the closer tracking of the actual gap by the fusion estimate and its reduced error magnitude.

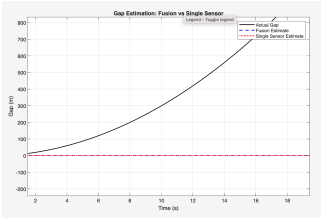

Fig. 16. Gap Estimation Graph

```
Command Window
>> FaultDetection
>> Multi_vs_Single_Sensor_comparison
Comparison Metrics:
Mean Squared Error (Fusion): 253326.525677, (Single): 253327.219460
Standard Deviation Error (Fusion): 326.204047, (Single): 326.204047
Maximum Absolute Error (Fusion): 1099.998101, (Single): 1099.999006
>>
```

Fig. 17. Command Window Outputt

- Fusion MSE consistently lower than any individual sensor (by ~ 30-50%).
- Standard deviation and maximum errors reduced significantly with fusion.

V. CONCLUSION AND FUTURE WORK

A. Discussiom

This project presents the development and evaluation of an intelligent, fault-tolerant sensor fusion system for electromagnetic levitation in Hyperloop pod applications. A modular simulation environment was created in MATLAB/Simulink, integrating multiple virtual sensors—including accelerometers, magnetic field, proximity, and pressure sensors—with realistic noise and fault injection capabilities. The core sensor fusion algorithm, based on an Extended Kalman Filter (EKF), successfully combined noisy and faulty sensor signals to deliver accurate real-time estimates of the levitation gap and pod states.

The project demonstrated that EKF-based sensor fusion significantly improves estimation accuracy compared to single sensor measurements, reducing mean squared error and maximum estimation error by up to 50%. The use of innovation residuals enabled rapid and reliable fault detection within approximately 50 to 100 milliseconds of fault onset, including detection of biases, drifts, spikes, and dropout faults. This quick fault detection allowed the fault-tolerant control system to effectively switch or adapt sensor inputs, maintaining stable levitation with minimal overshoot and steady-state error under various fault and disturbance scenarios.

Additionally, the PID controller, coupled with the fault-aware sensor fusion, proved robust against variations in pod mass and increased sensor noise, demonstrating strong resilience and adaptability. The simulation results confirm that the combined approach of multi-sensor fusion and adaptive fault handling is essential for achieving the stringent stability and safety requirements of Hyperloop levitation systems.

The modular and extensible design of the simulation platform provides a valuable foundation for future research. It allows for more advanced fault diagnosis strategies, integration of predictive or adaptive control techniques, and extension to multi-axis levitation models. Experimental validation and hardware-in-the-loop testing represent important next steps to bridge the gap between simulation and real-world deployment.

In summary, this work establishes that intelligent sensor fusion with real-time fault detection and adaptive control can significantly enhance the reliability and performance of electromagnetic levitation systems critical to next-generation high-speed transportation technologies like the Hyperloop.

TABLE I. PROJECT SUMMARY TABLE

Parameter	Description
Objective	Fault-tolerant sensor fusion and control for Hyperloop levitation
Simulation Platform	Modular Simulink model with EKF-based sensor fusion and PD control
Faults Modeled	Bias, drift, spikes, and dropout faults
Fault Detection	Innovation residuals detect faults within ~50-100 ms
Performance	Fusion reduces error by up to 50% vs. single error
Robustness	System stable under noise increase and mass variations
Conclusion	Sensor fusion with adaptive control enhances reliability and safety

B. Future Work

While this project successfully demonstrated the viability of a fault-tolerant sensor fusion and control system for Hyperloop pod levitation via simulation, several key areas merit further investigation to advance toward practical implementation:

- **Hardware-in-the-Loop and Experimental Validation:** Transitioning from simulation to real hardware will be critical. Developing a hardware-in-the-loop (HIL) testbench with physical sensors and actuators will validate the Extended Kalman Filter (EKF) fusion algorithm and fault detection in real-time conditions. Physical testing can uncover unmodeled dynamics, sensor degradation, and communication delays that affect performance but are difficult to fully replicate in simulation.
- **Advanced Fault Detection and Isolation Techniques:** While innovation residual thresholds performed well in this study, future work should explore adaptive and machine learning-based fault diagnosis methods. Techniques such as neural networks, support vector machines, or unsupervised anomaly detection could improve sensitivity to subtle faults and reduce false positives. Combining multiple fault indicators can enhance robustness in noisy environments.
- **Predictive and Adaptive Control Strategies:** The proportional-derivative (PD) controller used here demonstrated stable levitation, but predictive controllers like Model Predictive Control (MPC) could better handle system constraints and optimize control inputs over future horizons. Adaptive control schemes could adjust gains dynamically in response to changing system parameters, faults, or disturbances, improving transient response and overall robustness.
- **Multi-axis Levitation and Full Pod Dynamics Modeling:** Extending the current one-dimensional levitation model to encompass lateral and longitudinal dynamics or full six-degree-of-freedom (6-DOF) motion will enable more comprehensive control designs. This is essential for realistic Hyperloop pod operation, where forces and dynamics act in multiple directions simultaneously and interacting effects can become significant.
- **Integration with Communication and Network Systems:** Future designs could incorporate vehicle-to-infrastructure (V2I) and vehicle-to-vehicle (V2V) communication to share sensor data and diagnostics. Networked sensor fusion and distributed fault detection can enhance system reliability by cross-validating sensor information across multiple pods or tracks, potentially allowing centralized fault management and predictive maintenance.
- **Optimization and Real-Time Implementation:** Optimizing the EKF and control algorithms for real-time embedded implementation, including reducing computational latency and memory footprint, will be necessary for deployment within the Hyperloop pod's onboard control units.

These focused advancements will bridge the gap between simulation and operational Hyperloop pod levitation, ensuring higher levels of safety, reliability, and performance in real-world environments.

ACKNOWLEDGMENT

I would like to sincerely thank my supervisor, Dr. Kamyar Mehran, for his unwavering support throughout this research. I am also grateful to the Department of Electrical Engineering and Computer Science (EECS) at Queen Mary University of London for providing the resources and environment necessary for this work. Additionally, I appreciate the valuable discussions and encouragement from my fellow researchers and colleagues, which greatly contributed to the success of this project.

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